

30	A $A = \lambda N = \frac{\ln 2}{T_{1/2}} N$ where N is the number of molecules in the sample $N = n N_A$ And the A of each can be compared by finding the nuclide with the greatest ratio of $\frac{n}{T_{1/2}}$ Option A: $\frac{n}{T_{1/2}} = \frac{0.003}{10} = 0.0003$ Option B: $\frac{n}{T_{1/2}} = \frac{0.1}{400} = 0.00025$ Option C: $\frac{n}{T_{1/2}} = \frac{0.6}{2100} = 0.00029$ Option D: $\frac{n}{T_{1/2}} = \frac{1.0}{4800} = 0.00020$	A
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